

AI-Based Application for Task Management and Scheduling Student Activity

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Abstract—University students face significant challenges in managing academic demands, which often lead to procrastination, stress, and diminished mental well-being. To address this, we developed a proactive AI-based assistant designed to support student productivity and health. The application leverages Large Language Models (LLMs) and an agentic AI framework based on the ReAct pattern to offer personalized task prioritization, dynamic scheduling, and cognitive load reduction. It uniquely integrates academic, emotional, and biological factors, such as circadian rhythms, to provide holistic, context-aware support. A usability study involving 30 participants showed favorable outcomes, achieving a System Usability Scale (SUS) score of 73.67, a Task Success Rate (TSR) of 83% for the AI scheduling task, average Single Ease Question (SEQ) score of 5.61 on a 7-point scale (where higher is better) indicating that the system is good enough to use. Qualitative feedback highlighted user satisfaction with the system's stability and AI-driven scheduling capabilities. This research presents a novel, adaptive platform that shifts from reactive, siloed educational tools to an anticipatory support system. The findings validate the potential of agentic AI to enhance academic performance, offering a scalable model for future student support in higher education.

Keywords—agentic AI, educational technology, large language models (LLM), personalized scheduling, time management, SDG 3: good health and well-being, SDG 4: quality education.

I. INTRODUCTION

University life brings mounting academic demands, making effective time management a vital skill for students to support their well-being and achieve academic performance [1]. Without strong time management, many students fall into procrastination, with studies revealing that up to 60% of undergraduates regularly postpone their tasks, which is a direct cause of underachievement [2]. This cycle of pressure and poor time management takes a heavy toll on mental health. Students, particularly at the postgraduate level, face demands from their studies, family, and society that make them more vulnerable to develop anxiety and depression [3]. Adding to

this, a student's biological factor, such as natural sleep cycle or chronotype, also plays a role in their stress levels and classroom outcomes [4].

The consequences of these challenges are significant. One long-term study showed that students who struggled with mental health early in their studies were much more likely to achieve poor academic results throughout their entire time at university [5]. However, despite these issues being common, many students report a lack of guidance on how to manage the intense academic and psychological demands they face [6].

Conventional digital tools, such as calendars and to-do list applications, are inherently passive and reactive. They function like digital notepads, entirely dependent on manual input and lacking the situational awareness to anticipate future challenges [7]. This creates a critical disparity, forcing students to rely on non-adaptive digital aids for complex tasks, despite requiring dynamic, proactive support. To address this challenge, many state-of-the-art approaches in Large Language Models (LLMs) have given rise to systems that are capable of reasoning, planning, and acting autonomously. Such approach is called Agentic AI system. Based on this approach, we proposed an AI-based application, a proactive assistant to help students manage their tasks, schedule, and cognitive load. The main contribution of this work is its underlying context-aware platform architecture, specifically designed for a dynamic university environment.

II. RELATED WORKS

Educational technology is evolving from passive tools like digital calendars to proactive AI systems that address the multifaceted challenges of modern student life, including academic pressure, procrastination, and mental health [8]. AI is now pervasive in higher education, automating administrative tasks such as course scheduling and enrollment predictions, while providing 24/7 virtual assistants for academic and administrative support [9]. However, many systems remain fragmented, separating scheduling, tutoring, and mental health tools, thus ignoring the interconnectedness of academic stress and holistic well-being.

A. AI Personalization in Academic Scheduling and Task Management

Recent research on AI scheduling systems for students shows limitations in static inputs or narrow specialization. The following table compares related work, highlighting gaps in adaptability and holistic integration.

TABLE I. COMPARISON OF RESEARCH AI-BASED FOR STUDENT

Existing Study	Identified Research Gap
Personalized Student AI Scheduling Assistant [10].	Relies on static manual inputs, missing predictive adaptation for the dynamic school context.
Development of Dynamic Personalized Learning Paths Based on Knowledge Preferences and the Ant Colony Algorithm [11].	Limited to academic learning paths, lacking real-time adjustment for dynamic university life.
A Collaborative Control Protocol with Artificial Intelligence for Medical Student Work Scheduling [12].	Specific to medical students, neglecting broader school task management.
Artificial Intelligence Aided Course Scheduling System [13].	Focuses on course selection without integrating daily task prioritization or proactive AI for non-academic activities.
Large Language Models for Education: A Survey and Outlook [14]	Find a comprehensive survey on LLMs in education, but lacks implementation of an agentic AI framework for a proactive student approach.
Chatbots and Flipped Learning: Enhancing Student Engagement [15]	Focuses on chatbot engagement in flipped learning contexts but does not address personalized load scheduling.
Promoting Emotional and Psychological Well-being in Students through a Self-Help ChatGPT Intervention [16]	Specifically addresses mental health support through ChatGPT, but lacks integration with academic tasks.
Challenging Cognitive Load Theory: The Role of Educational Neuroscience and Artificial Intelligence in Redefining Learning Efficacy [17]	Systematic review of AI-driven adaptive learning for cognitive load optimization, but limited to instruction materials.
Integrating AI Chatbot Into LMS: Enhancing Student Engagement and Learning Outcomes [18]	Demonstrates chatbot effectiveness in LMS for engagement, but does not incorporate biological factors.
LLMs for Whole-Learner Support: Opportunities and Challenges [19]	Explores LLMs for cognitive and non-cognitive support but lacks a practical implementation framework.

B. Methodological Framework

To enable proactive, scalable AI applications in educational contexts, this work adopts a protocol for external integration called Modal Context Protocol (MCP), a standardized interface for agentic systems that facilitates dynamic interactions between LLMs and external resources. MCP ensures seamless integration of diverse resources, making it a very scalable framework. Additionally, the Reasoning and Acting (ReAct) framework is incorporated to enhance the capability of LLMs to perform complex tasks by interleaving reasoning with action-taking in external environments [20].

In order to make things run smoothly on everyday devices, the framework uses a client-server architecture. This setup tackles the limitation of mobile platforms by shifting heavy tasks, like LLM inference, to powerful central backends, all while keeping user experiences quick and responsive [21].

III. RESEARCH METHODOLOGY

Through informal discussions with students, the requirement analysis identified challenges in arranging tasks effectively and creating customized study plans, leading to the development of an AI-based application for proactive task scheduling. In response to these needs, an application capable of dynamically managing them was engineered with an intelligent assistant.

A. System Design

AI-based application is designed for proactive task scheduling and personalized recommendations. To achieve this, the system integrates a Large Language Model (LLM) with an Agentic AI framework as its foundational architecture, leveraging the MCP and ReAct Pattern. By adopting MCP, AI-based applications can dynamically exchange context with external data sources, such as calendars, thereby supporting adaptive and real-time scheduling [22].

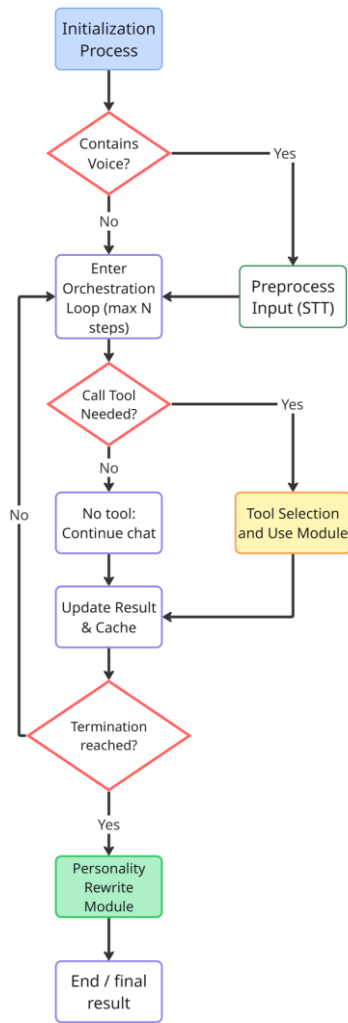


Fig. 1. Application Flow

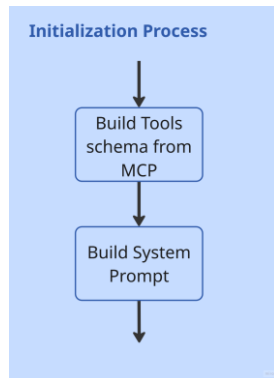


Fig. 2. Initialization Process

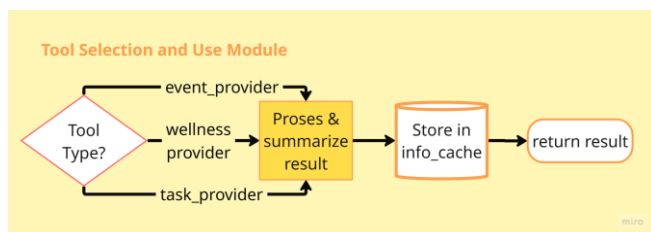


Fig. 3. Tool Selection and Use Modules

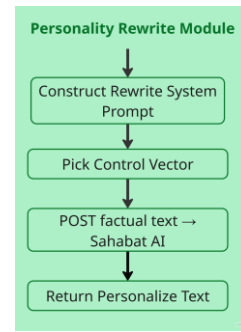


Fig. 4. Personalities Rewrite Modules

The system integrates representation learning to refine interaction output and improve the model's adaptability. Additionally, a Speech-to-Text (STT) module is incorporated to improve the user experience, utilizing the open-source caha/whisper-small model. The inclusion of STT is intended to lower interaction barriers in application use across diverse student contexts [23].

Through iterative prototyping, the interface was developed to prioritize intuitive navigation and a clear visual hierarchy, minimizing cognitive demands.

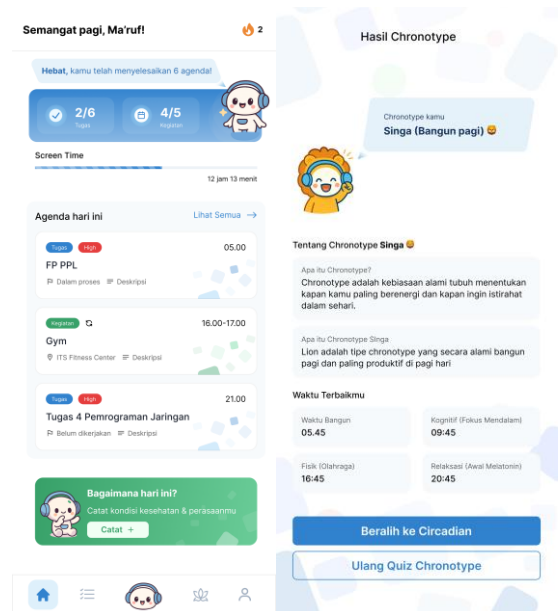


Fig. 5. Design User Interface

A primary color palette of blue and green was selected for its psychological benefits of promoting tranquility and focus [24]. Complementary and neutral tones were chosen to maintain accessibility, to WCAG 2.1 contrast guidelines for all users.

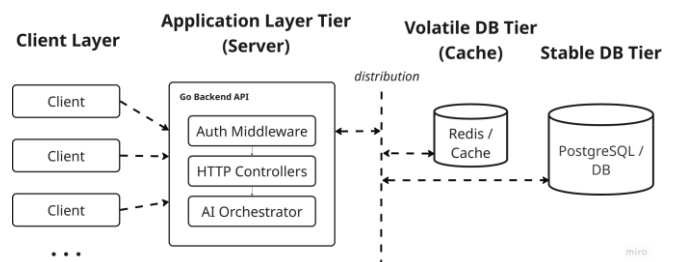


Fig. 6. System Architecture Client Server.

The application is deployed using a client-server architecture, comprising a mobile application as a client and a server-side AI processing backend. This configuration allows users to access computationally intensive processes, such as LLM or STT inference, with minimal resource consumption on their mobile devices. The choice of a client-server design is motivated by two factors: (i) resource efficiency, as mobile devices have limited capacity to execute LLM inference locally [25]; and (ii) contextual integration and scalability, since a centralized server can integrate external data sources, manage model updates, and maintain user profiles across sessions and devices [26]. This approach also improves scalability and reduces latency, consistent with findings from mobile-cloud and edge-centric architectures in prior research [27].

B. Evaluation

To assess the effectiveness of an AI-based application, both a task-based performance metric and a standardized usability assessment were employed. The System Usability Scale (SUS) was applied to measure the perceived usability of the application. The SUS is a well-established instrument in human-computer interaction research, calculated as:

$$SUS = 2.5 \times \sum_{i=1}^{10} S_i \quad (1)$$

Where S_i is the adjusted score for each of the ten questionnaire items, ranging from 0 to 4. The resulting SUS score ranges from 0 to 100, with higher values indicating greater usability [28]. A benchmark score of 68 was chosen for the study based on commonly cited literature. [29] Additionally, the Single Ease Question (SEQ) assesses the ease of completing specific tasks.

$$SEQ = \frac{\sum_{i=1}^U \sum_{j=1}^T r_{ij}}{U \cdot T} \quad (2)$$

Where r_{ij} is the rating given by user i for task j , U is the number of users, and T is the number of tasks. A higher SEQ score indicates greater ease of use, directly addressing cognitive overload and navigation challenges in the apps [30]. In addition, the Task Success Rate (TSR) was used to evaluate functional correctness. TSR measures the proportion of scheduled tasks successfully created by the AI agent, and is defined as:

$$TSR = \frac{1}{N} \sum_{i=1}^N S_i \quad (3)$$

Where N is the total number of tasks and $S_i = 1$ if the task was successfully scheduled, otherwise $S_i = 0$ [31]. This metric reflects the system's ability to generate accurate and functional schedules. By combining SUS, SEQ, and TSR, the evaluation framework validated system performance not only in terms of objective task execution but also subjective user experience quality.

IV. RESULT & DISCUSSION

A. Performance Evaluation

The evaluation was conducted with 30 students over 5 days, during which each participant completed 3 task scenarios related to specific modules (quiz and circadian,

agent AI, and Task Management). The Metrics indicators used to measure performance were SUS, SEQ, and TSR.

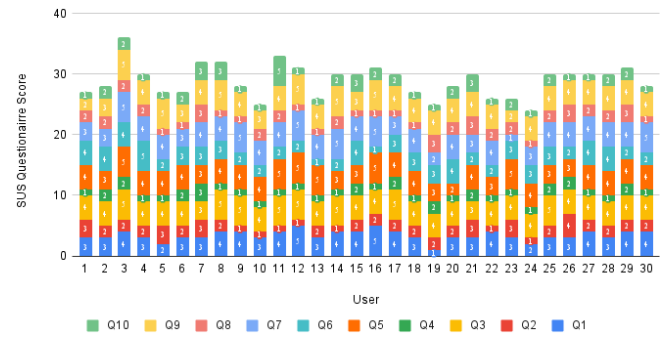


Fig. 7. SUS Questionnaire Test Result

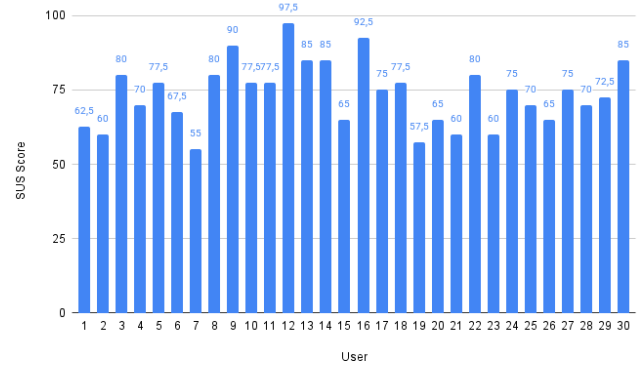


Fig. 8. SUS Score for Each User

The SUS score ranged from 55 to 97.5, with an average score of 73.67. This score lies above the acceptable usability benchmark mean of 68 commonly cited in literature, indicating that the app provides a moderately user-friendly interface. To gain more detailed understanding of usability across specific functionalities, SEQ was employed.

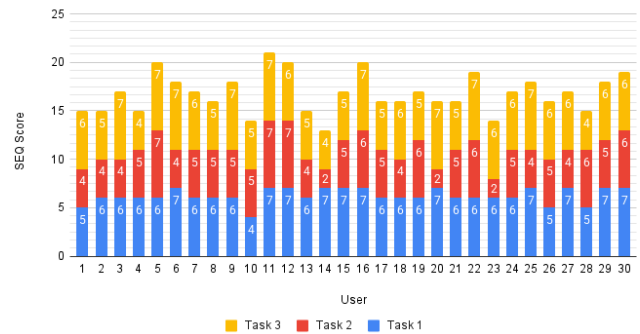


Fig. 9. SEQ Score per Task

With a total average SEQ score of 5.61 (on a scale of 1 to 7, where 1 is poor and 7 is excellent, with $n=7$), users rated about the ease of accessing quiz & circadian rhythms at 6.2 and Task Management at 5.83, reflecting a highly intuitive experience. On the other hand, the agent AI for scheduling scored lower at 4.8, indicating moderate difficulty, which may have come from a learning curve or complexity to customization. This variability in SEQ scores prompted further investigation into task success, leading to the analysis of TSR to assess the system's ability to support task completion effectively.

To ensure an objective and consistent evaluation, the TSR metric was operationalized with predefined success criteria for each task scenario. A task was considered a 'success' only if a user could complete it independently and achieve the correct final outcome without critical errors. Specifically, success criteria were: (1) completing the Circadian Quiz and viewing the results page, (2) having the AI agent generate an accurate, conflict-free schedule that the user accepted or saved, and (3) successfully adding and saving a new task with all required details in the Task Management module.

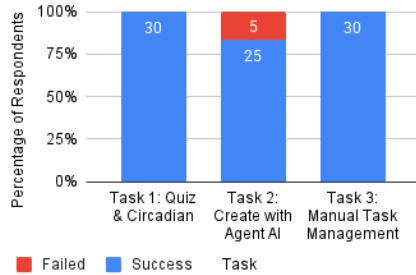


Fig. 10. TSR Score per Task

For Tasks 1 and 3, all users successfully completed the tasks, indicating a straightforward and user-friendly process. In contrast, Task 2 showed a lower completion rate, with 25 out of 30 users succeeding, which corresponds to 83%. To gain deeper insights into these challenges, user feedback was collected and analyzed.

TABLE II. USER FEEDBACK

Theme	User Comments
Design Quality	"Logo and design assets are not yet HD."
Onboarding Guidance	"Please create a feature tour as I was confused at the start about how to use the app."
Frontend Display Issue	Text in the frontend still overlaps
Prompt Examples	"Please provide example prompts at the start for better guidance."
System Stability	"Good Experience with no bugs"
AI Formatting	"AI formatting still needs improvement."
AI Hallucination	"The AI Agent sometimes hallucinates and forgets messages."
AI Response Time	"The AI output is not fast enough."
Calendar and Task overview	"Need to add a monthly date overview in the frontend to easily view tasks by month."

Qualitative feedback highlighted key themes in Table II, encompassing both positive and negative feedback, identifying challenges and suggesting areas for improvement in order to enhance usability and reduce variability in TSR.

B. Discussion

The SUS score of 73.67 exceeds the 68.5 benchmark. The average SEQ scores are 6.2 for quiz/circadian, 4.8 for AI scheduling, and 5.6 for task management, highlighting strength in circadian & task management features, but low AI scheduling score reflecting feedback on moderate difficulty. The analysis of Task Success Rate (TSR) provides deeper insight into the usability of each feature. All users successfully completed Task 1 and Task 3, but only 83% users were able to complete Task 2. This indicates that the AI scheduling feature imposes additional complexity and may not yet be as accessible or reliable as the other modules. To address these challenges, future enhancements will focus on increasing the reliability and user guidance within the AI scheduler. Planned improvements include developing more structured input methods, introducing templated output formats to reduce ambiguity and formatting errors, and adding a mandatory review and confirmation step for users before finalizing AI-generated schedules. Additionally, enhancing the onboarding process with interactive tours and effective prompt examples is expected to mitigate the learning curve and address specific difficulties highlighted by users. This is directly supported by user feedback, such as "The AI Agent sometimes hallucinates and forgets messages" and "AI formatting still needs improvement." This unpredictability leads to a steeper learning curve and can cause user frustration, contributing to lower success rates and perceived ease of use for some participants.

V. CONCLUSION

In this research, we addressed the challenges university students face in managing academic demands by developing an AI-based application that functions as a proactive assistant. Leveraging LLMs and agent AI frameworks, the system is designed to prioritize tasks and create personalized schedules to reduce cognitive load. An evaluation with 30 participants yielded promising results, with an acceptable SUS score of 73.67, an attained 5.61 SEQ score, and a perfect task completion rate for quiz and manual task features. Meanwhile, the AI scheduling feature reached a task success rate of 83%, highlighting both its potential and its current limitations.

The primary contribution of this work is a holistic and adaptive platform that bridges the gap in existing educational AI tools. Unlike static schedulers, our application proactively anticipates students' needs by integrating cognitive load tasks and activities, emotional logs, and biological factors to improve both productivity and well-being. This comprehensive approach directly supports SDG 3 (Good Health and Well-being) by prioritizing students' mental health, reducing academic stress, and promoting sustainable learning practices that prevent burnout. Future development will focus on mitigating AI hallucinations and simplifying the interface.

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